

Project Initialization and Planning Phase

Date	29 June 2026
Team ID	
Project Title	OptiCrop – Smart Agricultural Optimization Engine
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed solution aims to improve agricultural productivity using smart technologies and data-driven decision-making. This system helps farmers make better decisions regarding crop selection, irrigation, and resource utilization.

Project Overview	
Objective	The main objective is to develop an intelligent system that assists farmers in optimizing crop yield by analyzing environmental and soil data.
Scope	The project focuses on improving agricultural efficiency using data analytics, sensor inputs, and predictive models to guide farming decisions.
Problem Statement	
Description	Traditional farming methods lack precision and often result in poor yield due to improper decision-making regarding crops, irrigation, and fertilizers.
Impact	Solving these issues will increase crop productivity, reduce resource wastage, and improve farmers' income and sustainability.
Proposed Solution	
Approach	The system uses data analysis and machine learning techniques to provide recommendations on crop selection, irrigation scheduling, and soil management.
Key Features	• Smart crop recommendation system • Soil and weather data analysis • Real-time decision support • Efficient resource management

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	T4 GPU
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	1 TB SSD
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy, matplotlib, seaborn
Development Environment	IDE	Jupyter Notebook, pycharm
Data		
Data	Agricultural datasets (soil, weather, crop yield)	CSV